

flowchart LR

```
/ekf_filter_node[ /ekf_filter_node ]:::main
/fan_controller[ /fan_controller ]:::main
/front_camera/camera_node[ /front_camera/camera_node ]:::main
/line_follow/navigation[ /line_follow/navigation ]:::main
/oled_controller[ /oled_controller ]:::main
/servo_lift[ /servo_lift ]:::main
/transform_listener_impl_555586bbff30[ /
transform_listener_impl_555586bbff30 ]:::main
/state_machine[ /state_machine ]:::node
/pwm2/pwm_controller[ /pwm2/pwm_controller ]:::node
/line_follow/navigation[ /line_follow/navigation ]:::node
/line_follow/pid_loop[ /line_follow/pid_loop ]:::node
/i2c_controller[ /i2c_controller ]:::node
/imu/data([ /imu/data<br>sensor_msgs/msg/Imu ]):::bugged
/set_pose([ /set_pose<br>geometry_msgs/msg/
PoseWithCovarianceStamped ]):::bugged
/wheel/odom([ /wheel/odom<br>nav_msgs/msg/Odometry ]):::bugged
/accel/filtered([ /accel/filtered<br>geometry_msgs/msg/
AccelWithCovarianceStamped ]):::bugged
/diagnostics([ /diagnostics<br>diagnostic_msgs/msg/
DiagnosticArray ]):::bugged
/odometry/filtered([ /odometry/filtered<br>nav_msgs/msg/
Odometry ]):::bugged
/fan/target_speed([ /fan/target_speed<br>std_msgs/msg/
Int32 ]):::topic
/fan/speed([ /fan/speed<br>std_msgs/msg/Int32 ]):::bugged
/pwm2/duty_cycle([ /pwm2/duty_cycle<br>std_msgs/msg/
Int32 ]):::topic
/pwm2/enable([ /pwm2/enable<br>std_msgs/msg/Bool ]):::topic
/pwm2/period([ /pwm2/period<br>std_msgs/msg/Int32 ]):::topic
/front_camera/camera_node/camera_info([ /front_camera/camera_node/
camera_info<br>sensor_msgs/msg/CameraInfo ]):::bugged
/front_camera/camera_node/image_raw([ /front_camera/camera_node/
image_raw<br>sensor_msgs/msg/Image ]):::topic
/front_camera/camera_node/image_raw/compressed([ /front_camera/
camera_node/image_raw/compressed<br>sensor_msgs/msg/
CompressedImage ]):::bugged
/line_follow/line_error([ /line_follow/line_error<br>std_msgs/msg/
Float64 ]):::topic
/line_error([ /line_error<br>std_msgs/msg/Float32 ]):::bugged
/oled_black([ /oled_black<br>std_msgs/msg/Int32 ]):::bugged
/oled_silver([ /oled_silver<br>std_msgs/msg/Int32 ]):::bugged
/oled_status([ /oled_status<br>std_msgs/msg/String ]):::bugged
/stm_temp([ /stm_temp<br>std_msgs/msg/Float32 ]):::topic
/servo/lift([ /servo/lift<br>std_msgs/msg/Float32 ]):::bugged
/ekf_filter_node/get_type_description[/ /ekf_filter_node/
get_type_description<br>type_description_interfaces/srv/
GetTypeDescription \]:::bugged
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/enable[/ /enable<br>std_srvs/srv/Empty \]:::bugged
/reset[/ /reset<br>std_srvs/srv/Empty \]:::bugged
/set_pose[/ /set_pose<br>robot_localization/srv/SetPose
\]:::bugged
/toggle[/ /toggle<br>robot_localization/srv/ToggleFilterProcessing
\]:::bugged
/fan_controller/get_type_description[/ /fan_controller/
get_type_description<br>type_description_interfaces/srv/
GetTypeDescription \]:::bugged
/front_camera/camera_node/get_type_description[/ /front_camera/
camera_node/get_type_description<br>type_description_interfaces/
srv/GetTypeDescription \]:::bugged
/front_camera/camera_node/set_camera_info[/ /front_camera/
camera_node/set_camera_info<br>sensor_msgs/srv/SetCameraInfo
\]:::bugged
/line_follow/navigation/get_type_description[/ /line_follow/
navigation/get_type_description<br>type_description_interfaces/
srv/GetTypeDescription \]:::bugged
/oled_controller/get_type_description[/ /oled_controller/
get_type_description<br>type_description_interfaces/srv/
GetTypeDescription \]:::bugged
/servo_lift/get_type_description[/ /servo_lift/
get_type_description<br>type_description_interfaces/srv/
GetTypeDescription \]:::bugged
/i2c_write[/ /i2c_write<br>robot_msgs/srv/I2CWrite \]:::service
/transform_listener_impl_555586bbff30/get_type_description[/ /
transform_listener_impl_555586bbff30/
get_type_description<br>type_description_interfaces/srv/
GetTypeDescription \]:::bugged

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/imu/data --> /ekf_filter_node
/set_pose --> /ekf_filter_node
/wheel/odom --> /ekf_filter_node
/fan/target_speed --> /fan_controller
/front_camera/camera_node/image_raw --> /line_follow/navigation
/line_error --> /oled_controller
/oled_black --> /oled_controller
/oled_silver --> /oled_controller
/oled_status --> /oled_controller
/stm_temp --> /oled_controller
/servo/lift --> /servo_lift
/pwm2/duty_cycle --> /pwm2/pwm_controller
/pwm2/enable --> /pwm2/pwm_controller
/pwm2/period --> /pwm2/pwm_controller
/front_camera/camera_node/image_raw --> /line_follow/navigation
/line_follow/line_error --> /line_follow/pid_loop
/ekf_filter_node --> /accel/filtered
/ekf_filter_node --> /diagnostics
/ekf_filter_node --> /odometry/filtered
/fan_controller --> /fan/speed
/fan_controller --> /pwm2/duty_cycle
/fan_controller --> /pwm2/enable
/fan_controller --> /pwm2/period

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/front_camera/camera_node --> /diagnostics
/front_camera/camera_node --> /front_camera/camera_node/
camera_info
/front_camera/camera_node --> /front_camera/camera_node/image_raw
/front_camera/camera_node --> /front_camera/camera_node/image_raw/
compressed
/line_follow/navigation --> /line_follow/line_error
/state_machine --> /fan/target_speed
/i2c_controller --> /stm_temp
/ekf_filter_node/get_type_description o-.-o /ekf_filter_node
/enable o-.-o /ekf_filter_node
/reset o-.-o /ekf_filter_node
/set_pose o-.-o /ekf_filter_node
/toggle o-.-o /ekf_filter_node
/fan_controller/get_type_description o-.-o /fan_controller
/front_camera/camera_node/get_type_description o-.-o /
front_camera/camera_node
/front_camera/camera_node/set_camera_info o-.-o /front_camera/
camera_node
/line_follow/navigation/get_type_description o-.-o /line_follow/
navigation
/oled_controller/get_type_description o-.-o /oled_controller
/servo_lift/get_type_description o-.-o /servo_lift
/transform_listener_impl_555586bbff30/get_type_description o-.-o /
transform_listener_impl_555586bbff30
/i2c_write o-.-o /i2c_controller
/servo_lift <-.-> /i2c_write

```

```

subgraph keys[<b>Keys<b/>]
subgraph nodes[<b><b/>]
topicb((No connected))::bugged
main_node[main]::main
end
subgraph connection[<b><b/>]
node1[node1]::node
node2[node2]::node
node1 o-.-o|to server| service[/Service<br>service/
Type\]::service
service <-.->|to client| node2
node1 -->|publish| topic([Topic<br>topic/Type])::topic
topic -->|subscribe| node2
node1 o==o|to server| action[{/Action<br>action/Type/}]::action
action <==>|to client| node2
end
end
classDef node opacity:0.9,fill:#2A0,stroke:#391,stroke-
width:4px,color:#fff
classDef action opacity:0.9,fill:#66A,stroke:#225,stroke-
width:2px,color:#fff
classDef service opacity:0.9,fill:#3B8062,stroke:#3B6062,stroke-
width:2px,color:#fff
classDef topic opacity:0.9,fill:#852,stroke:#CCC,stroke-

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width:2px,color:#fff
classDef main opacity:0.9,fill:#059,stroke:#09F,stroke-
width:4px,color:#fff
classDef bugged opacity:0.9,fill:#933,stroke:#800,stroke-
width:2px,color:#fff
style keys opacity:0.15,fill:#FFF
style nodes opacity:0.15,fill:#FFF
style connection opacity:0.15,fill:#FFF
```